ENGINE [SKYACTIV-G 1.3, SKYACTIV-G 1.5]

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Special Features

- For SKYACTIV-G 1.3 and SKYACTIV-G 1.5 are the following have been adopted to lower fuel consumption.
  - — Mechanical resistance loss improvement from:
    - Narrowed down crankshaft journal
    - Optimized piston skirt shape
    - Lowered piston ring tension
    - Reduction of valve spring load
    - Stabilization of timing chain behavior
    - Optimized engine coolant passage
    - Optimized water pump impeller shape
    - Lowered drive belt tension
    - Optimized oil passage
    - Optimized oil pump shape
    - Oil pump discharging pressure control has been adopted. (SKYACTIV-G 1.5 (with 4-2-1 exhaust system))
  - — Pumping loss improvement from:
    - Variable valve timing mechanism adopted on both sides of intake and exhaust. (SKYACTIV-G 1.5 (with 4-2-1 exhaust system))
    - Variable valve timing structure on intake side adopted (SKYACTIV-G 1.5 (with 4-1 exhaust system))
  - — Cooling loss improvement
    - Piston cavity has been adopted.
  - — Combustion efficiency improvement
    - Multiple hole-type fuel injectors adopted
- L-jetronic*1 and D-jetronic*2 type detectors have been combined for intake air amount detection, improving the accuracy of the intake air amount measurement.
  - — MAF sensor adopted
  - — MAP sensor adopted
  - — IAT sensor No.1 and No.2 adopted
- Valve timing control has been adopted on both sides of the intake and exhaust, improving fuel economy and emission performance. (SKYACTIV-G 1.5 (with 4-2-1 exhaust system))
  Intake side: Electric variable valve timing control
  - — Intake CMP sensor adopted
  - — Electric variable valve timing motor/driver adopted
  - — Electric variable valve timing relay adopted

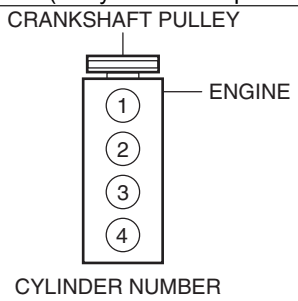
  Exhaust side: Hydraulic variable valve timing control
  - — Exhaust CMP sensor adopted
- Engine oil control has been adopted reducing engine load. (SKYACTIV-G 1.5 (with 4-2-1 exhaust system))
  - — Engine oil solenoid valve adopted
- With the adoption of fuel pump control, fuel pump power consumption has been reduced, improving fuel economy.
  - — Fuel pump control module adopted
- Generator output control adopted, fuel economy/idling stability improved.
  - — Current sensor adopted
- DC-DC converter control has been adopted for improved power supply stability. (Without i-ELOOP)
  - — DC-DC converter adopted
- With the adoption of the i-ELOOP, charging efficiency during deceleration is improved. (See i-ELOOP [i-ELOOP].) (With i-ELOOP)
  - — DC-DC converter (i-ELOOP) adopted
- With the adoption of the ion sensor, which detects pre-ignition, engine reliability has been improved. (SKYACTIV-G 1.5 (with 4-2-1 exhaust system))
- LIN communication has been adopted to the current sensor and DC-DC converter for simplified wiring harnesses. (Without i-ELOOP)
- i-stop control has been adopted for improved fuel efficiency, reduced exhaust gas emissions, and reduced idling noise. (With i-stop control)

*1 : Measures the intake air amount directly using the MAF sensor.

*2 : Measures the intake air pressure introduced into the cylinder using the MAP sensor and calculates the intake air amount indirectly.

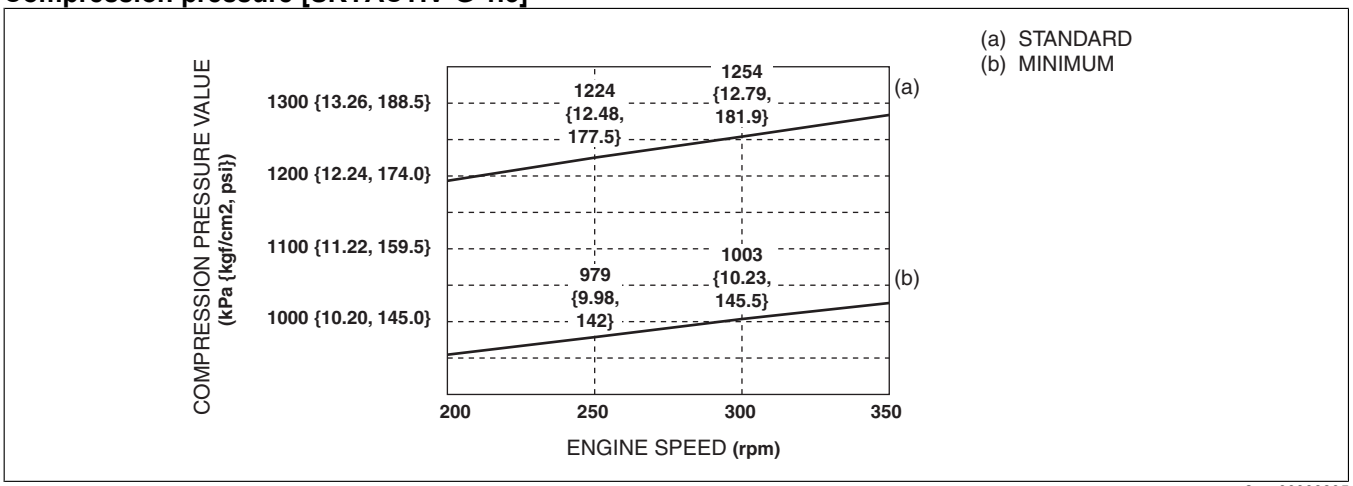
Specification

Item				Specification		
				SKYACTIV-G 1.3	SKYACTIV-G 1.5 (with 4-1 exhaust system)	SKYACTIV-G 1.5 (with 4-2-1 exhaust system)
MECHANICAL						
Type				Gasoline, 4-cycle		
Cylinder arrangement and number				In-line, 4-cylinder		
Combustion chamber				Pentroof		
Valve system				DOHC, timing chain driven, 16 valves		
Displacement			(ml {cc})	1,298 {1,298}	1,496 {1,496}	
Bore × stroke			(mm {in})	71.0 × 82.0 {2.80 ×3.23}	74.5 × 85.8 {2.93 × 3.38}	
Compression ratio				12.0:1		**Piston set part number: P5Y2-11SA0** 13.0:1 **Piston set part number: P5Y1-11SA0** 14.0:1
Valve timing	IN	Open	(°)	BTDC 43—ATDC 8		BTDC 38—ATDC 36
		Close	(°)	ABDC 31—ABDC 82		ABDC 36—ABDC 110
	EX	Open	(°)	BBDC 49		BBDC 59—BBDC 14
		Close	(°)	ATDC 5		ATDC 5—ATDC 50
LUBRICATION SYSTEM						
Type				Force-fed type		
Oil pressure (reference value) [Water temperature: 80—90 °C {176—194 °F}]			(kPa {kgf/cm^2, psi} [rpm])	230—400 {2.35—4.07, 33.4—58.0} [3,000]		Low oil; pressure: 110—175 {1.13—1.78, 16.0—25.3} [1,500] High oil pressure: 300—430 {3.06—4.38, 43.6—62.3} [4,500]
Oil pump	Type			Trochoid gear type		
Oil filter	Type			Full-flow, paper element		
	Bypass pressure		(kPa {kgf/cm^2, psi})	78—118 {0.80—1.20, 12.0—17.1}		
Oil capacity (approx. quantity)	Total (dry engine)		(L {US qt, Imp qt})	4.7 {5.0, 4.1}		
	Oil replacement		(L {US qt, Imp qt})	3.9 {4.1, 3.4}		
	Oil and oil filter replacement		(L {US qt, Imp qt})	4.2 {4.4, 3.7}		
COOLING SYSTEM						
Type				Water-cooled, Electromotive		
Water pump	Type			Centrifugal, V-ribbed belt-driven		
Thermostat	Type			Wax type		
	Opening temperature		(°C {°F})	86.5—89.5 {188—193}		
	Full-open temperature		(°C {°F})	100 {212}		
	Full-open lift		(mm {in})	8.5 {0.33} or more		
Radiator	Type			Corrugated fin type		
Cooling system cap	Valve opening pressure		(kPa {kgf/cm^2, psi})	93.2—122.6 {0.951—1.250, 13.6—17.7}		
Cooling fan type A	Type			Electric type		
	Number of blades			7		
	Outer diameter		(mm {in})	340 {13.4}		
	Cooling fan motor output		(W)	120		

Item			Specification		
			SKYACTIV-G 1.3	SKYACTIV-G 1.5 (with 4-1 exhaust system)	SKYACTIV-G 1.5 (with 4-2-1 exhaust system)
Cooling fan type B	Type		Electric type		
	Number of blades		9		
	Outer diameter	(mm {in})	340 {13.4}		
	Cooling fan motor output	(W)	120		
FUEL SYSTEM					
Fuel injector	Type		High resistance		
	Fuel supply method		Top-feed		
	Drive types		Electronic type		
Pressure regulator control pressure		(kPa {kgf/cm^2, psi})	European (L.H.D.), General (L.H.D.) specs.: Approx. 430 {4.38, 62.4} Except European (L.H.D.), General (L.H.D.) specs.: Approx. 450 {4.59, 65.3}		
Fuel pump	Type		Electric		
Fuel tank	Capacity	(L {US gal, Imp gal})	Thailand: 35 {37.0, 31.0} Except Thailand: 44 {46.0, 39.0}		
EMISSION SYSTEM					
Catalyst	Type		WU-TWC (monolith) TWC (monolith)		
EVAP control system	Type		Charcoal canister type		
PCV system	Type		Closed type		
CHARGING SYSTEM					
Battery	Voltage	(V)	12		
	Type (with i-stop)		Q-85		Without i-ELOOP: Q-85 With i-ELOOP: N-55
	Type and capacity (without i-stop)	(A·h/20HR)	—	55D23L (60), GR35 (60)	Without i-ELOOP: GR35 (60) With i-ELOOP: N-55
Generator	Output	(V-A)	12-100		Without i-ELOOP: 12-100 With i-ELOOP: 12-150
IGNITION SYSTEM					
Ignition system	Type		SEI		
	Spark advance		Electronic		
	Firing order		1—3—4—2 (all cylinders independent firing)		
					
Spark plug	Type		P51R-18-110		PE5R-18-110 PE5S-18-110
STARTING SYSTEM					
Starter	Type		Coaxial reduction		
	Output	(kW)	1.4	Without i-stop: 1.0 With i-stop: 1.4	
CONTROL SYSTEM					
i-stop OFF switch			ON/OFF		
Neutral switch			ON/OFF		
CPP switch			ON/OFF		
MAF sensor			Hot film		
IAT sensor			Cold resistor		
MAP sensor			Sealant diaphragm		
TP sensor			Hall element		
APP sensor			Electromagnetic inductiont		

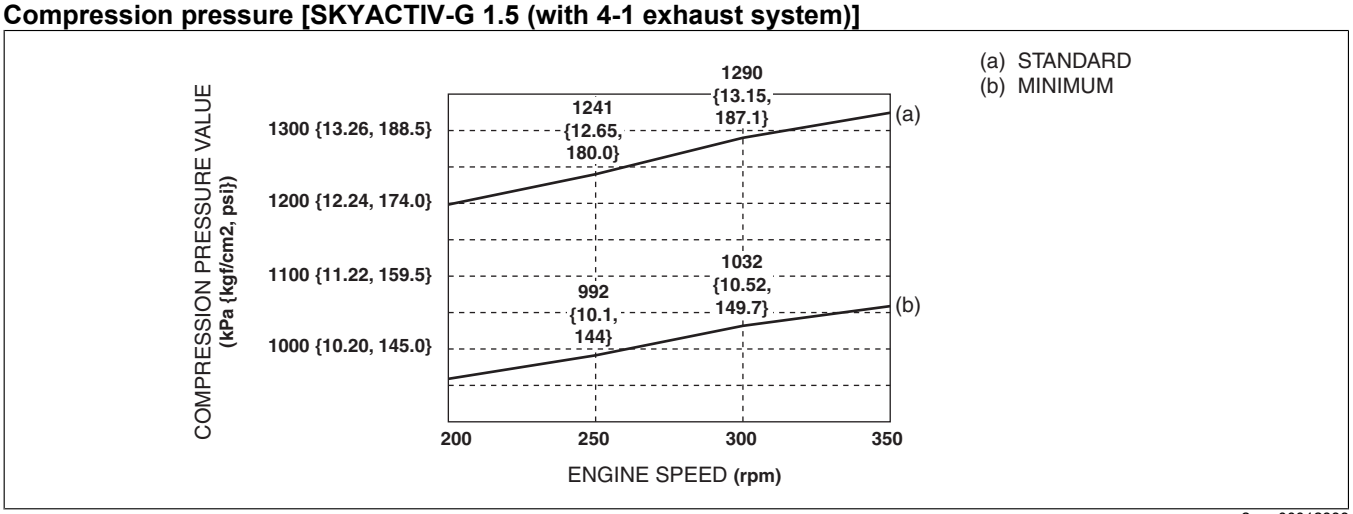
Item	Specification		
	SKYACTIV-G 1.3	SKYACTIV-G 1.5 (with 4-1 exhaust system)	SKYACTIV-G 1.5 (with 4-2-1 exhaust system)
CKP sensor	MR element		
CMP sensor	GMR element		
ECT sensor	Thermistor		
BARO sensor	Piezoelectric element		
Fuel pressure sensor	Metal diaphragm		
KS	Piezoelectric element		
Current sensor	Shunt resistance, Thermistor		
A/F sensor	Zirconium element		
HO2S	Zirconia element		
Clutch stroke sensor	Hall element		
Power brake unit vacuum sensor	Piezoelectric element		

Compression pressure [SKYACTIV-G 1.3]



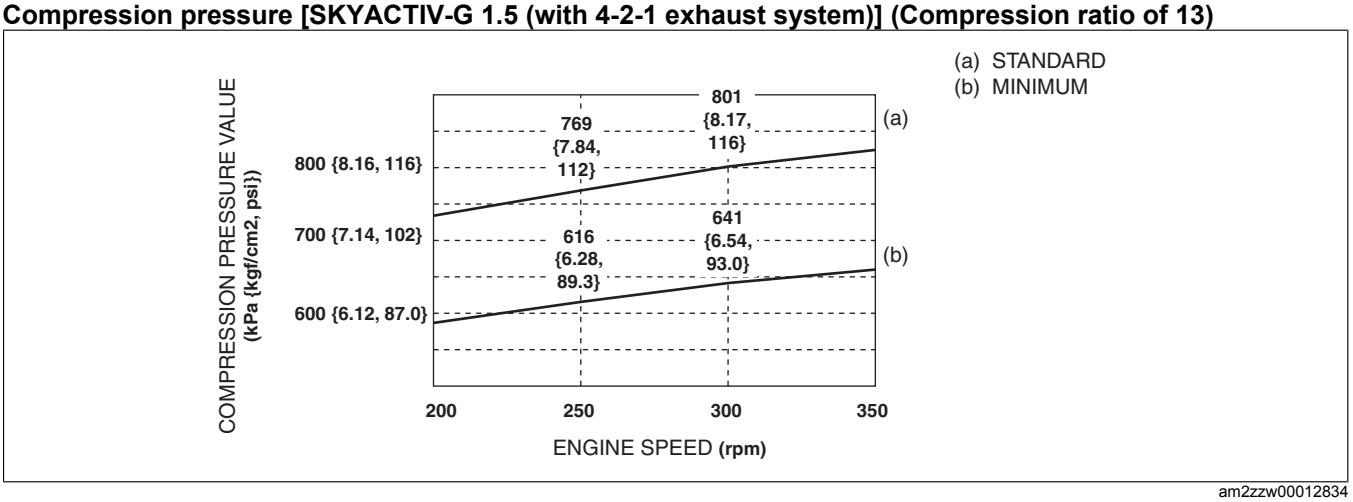
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Compression pressure [SKYACTIV-G 1.5 (with 4-1 exhaust system)]



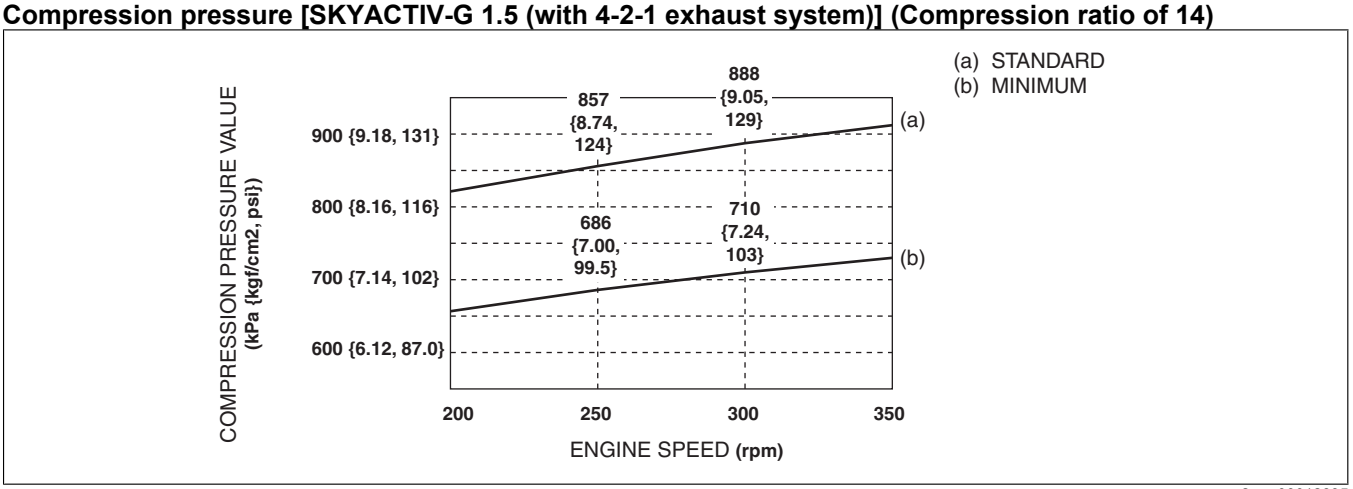
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Compression pressure [SKYACTIV-G 1.5 (with 4-2-1 exhaust system)] (Compression ratio of 13)



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Compression pressure [SKYACTIV-G 1.5 (with 4-2-1 exhaust system)] (Compression ratio of 14)



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Engine oil [Australian specs.]

Item	Specifications
Grade	API SG/SH/SJ/SL/SM/SN or ILSAC GF-2/GF-3/GF-4/GF-5
Viscosity (SAE)	10W-30, 5W-20, 5W-30, 0W-20, 0W-30

Engine oil [European (L.H.D. U.K.) specs.]

Recommended engine oil		Alternative engine oil
Mazda Original Oil Supra 0W-20	Mazda Original Oil Ultra 5W-30	API SL/SM/SN or ACEA A3/B3 or A5/B5 0W-20, 5W-30

Engine oil [Except Australian, European (L.H.D. U.K.), specs.] (SKYACTIV-G 1.3, SKYACTIV-G 1.5 (with 4-1 exhaust system))

Item	Specifications
Grade	API SG/SH/SJ/SL/SM/SN*1 or ILSAC GF-2/GF-3/GF-4/GF-5
Viscosity (SAE)	10W-30, 5W-20, 5W-30, 0W-20, 0W-30

*1 : For African nations, use SL or higher engine oil with the SKYACTIV-G 1.3 and SKYACTIV-G 1.5 (with 4-1 exhaust system) engines.

Engine oil [Except Australian, European (L.H.D. U.K.), specs.] (SKYACTIV-G 1.5 (with 4-2-1 exhaust system))

Item	Specifications
Grade	API SG/SH/SJ/SL/SM/SN or ILSAC GF-2/GF-3/GF-4/GF-5
Viscosity (SAE)	10W-30, 10W-40, 10W-50, 5W-20, 5W-30, 5W-40, 0W-20, 0W-30

Engine coolant capacity (approx. quantity)

Item		Capacity
European (L.H.D.), General (L.H.D.), Russia specs.	MTX	5.4 L {5.7 US qt, 4.8 Imp qt}
	ATX	5.6 L {5.9 US qt, 4.9 Imp qt}

Item			Capacity
Australian, European (R.H.D.) specs.		MTX	5.0 L {5.3 US qt, 4.4 Imp qt}
		ATX	5.2 L {5.5 US qt, 4.6 Imp qt}
General (R.H.D.) specs.(SKYACTIV-G 1.3)			
General (R.H.D.) specs. (SKYACTIV-G 1.5)	MTX	Without heater	4.5 L {4.8 US qt, 4.0 Imp qt}
		With heater	5.0 L {5.3 US qt, 4.4 Imp qt}
	ATX	Without heater	4.7 L {5.0 US qt, 4.1 Imp qt}
		With heater	5.2 L {5.5 US qt, 4.6 Imp qt}
Thailand specs.	Without heater		4.9 L {5.2 US qt, 4.3 Imp qt}
	With heater		5.4 L {5.7 US qt, 4.8 Imp qt}

Fuel type

Fuel	Research Octane Number	Country
Premium unleaded fuel (Conforming to EN 228 and within E10)*1	95 or above	Cyprus, Malta, Ireland, UK, New Caledonia, Turkey, Azerbaijan, Armenia, Georgia, Kazakhstan, Canary Islands, Reunion, Morocco, Austria, Greece, Italy, Switzerland, Belgium, Denmark, Finland, Norway, Portugal, Spain, Sweden, Hungary, Germany, Poland, Bulgaria, Croatia, Slovenia, Luxembourg, Slovakia, Latvia, Lithuania, Russia, Belarus, France, Ukraine, Czech, Estonia, Faeroe, Iceland, Romania, The Netherlands, Macedonia, Bosnia And Herzegovina, Serbia, Montenegro, Albania, Moldova, Martinique, F. Guiana, Guadeloupe, Tahiti, Vanuatu, Lebanon, Israel, Algeria, Tunisia, Madagascar, Guatemala, Bolivia, Uruguay, Honduras, Nicaragua, Aruba, St. Martin, UAE, Taiwan, Libya, Singapore, Hong Kong, Macau, Malaysia, Brunei, Mauritius, Seychelles, South Africa, Bermuda, Jamaica, Barbados, Antigua
Premium unleaded fuel (within E20)	95 or above	Thailand
Regular unleaded fuel (Conforming to Fuel Quality Standards Act 2000)*2	92 or above	Egypt, Myanmar
	90 or above	Australia, New Zealand, The Philippine, Cambodia, Laos, Kuwait, Oman, Qatar, Saudi Arabia, Syria, Bahrain, Jordan, Cameroon, Ivory Coast, Nigeria, Angola, Burundi, Gabon, Ghana, Chile, El Salvador, Costa Rica, Ecuador, Haiti, Colombia, Dominican Republic (LHD), Panama, Peru, B. Virgin, Curacao, Indonesia, Nepal, Sri Lanka, Fiji, Kenya, Tanzania, Mozambique, Trinidad and Tobago, Vietnam

*1 : Europe

*2 : Australia